Sheet (1)

(1) Write a program that displays the following four lines;

Electrical Engineering Department

Program: Electrical Power and Machines Engineering

Course: Computer Language and Applications

Code: GEN 241

(2) Write a program that displays Welcome to C++ five times.

(3) Write a program that displays the following pattern:

```
*          *********
***         *******
*****        *****
*******        ***
*********        *
```

(4) Write a program that displays the following table:

P	P*5	P*10
5	25	50
10	50	100
25	125	250
50	250	500

(5) Write a program that displays the result of $\dfrac{1.2\times0.1+3.3\times0.3}{0.09+0.001}$

(6) Write a program that displays the sum of the first ten odd numbers.

(7) an approximate value of π can be computed by the following formula:

$$\pi = 4\left(1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\frac{1}{9}-\frac{1}{11}+\frac{1}{13}-\frac{1}{15}\right)$$

Use 1.0 instead of 1 in your program and discuss why the result is different.

(8) Write a program that displays the area and perimeter of an equilateral triangle that has its three sides as 9.2, using the following formula:

$$area = 1.732 \times (side)^2/4$$

$$perimeter = 3 \times side$$

(9) Write a program that displays the area and perimeter of a square that has a side of 5.2 using the following formula:

$$area = (side)^2 \quad and \quad perimeter = 4 \times side$$

(10) Assume a vendor sells 600 kilograms of grocery in 15 hours, 30 minutes and 30 seconds. Write a program that displays the average sale in grams per hour.

(11) Assume a vendor sells 55.53 kilograms of grocery in 2 hours, 9 minutes and 30 seconds. Write a program that displays the average sale in kilograms per hour.

(12) According to population statistics in Egypt at the end of 2025, there is:

- There are 226.625 births per hour
- There are 70.583 deaths per hour

Assume that the population at the end of 2025 is 118,058,996 and one year has 365 days. Write a C++ program to estimate and display the population for each of the next five years.